Neguse Solomon Mekonnen

Neguse Solomon Mekonnen

Master's Degree in Civil EngineeringGraduate Researcher at Polytechnic University of Catalonia

Barcelona, Spain

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About me

I am a civil and software engineer and researcher focused on infrastructure risk, structural reliability, and data-driven engineering. Holding an Erasmus Mundus Master's (NoRISK) from Padua and UPC Barcelona, my work spans bridge reliability using Bayesian inference and deep-learning surrogate twins for urban drainage. Currently completing a CS degree at UoPeople, I combine engineering and programming to build resilient systems and am eager to collaborate with fellow researchers.

Disciplines

Civil Engineering

Structural Engineering

Software Engineering

Artificial Intelligence

Contact information

Website

Research Spotlight

Featured research

Deep-Learning Surrogate Digital Twins for Predictive Capacity Auditing and Maintenance Optimization

Preprint

November 2026

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Affiliations

Polytechnic University of Catalonia

Polytechnic University of Catalonia

Barcelona, Spain

Primary

March 2026

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Present

Graduate Researcher

ETSECCPB - School of Civil Engineering of Barcelona

Applied Bayesian MCMC to monitoring data from the 2015 proof load test of the ASR-deteriorated Zijlweg Viaduct. Used C++ and Latin Hypercube Sampling to model structural response, quantified a +1.85 reliability-index benefit from monitoring data, and assessed safety

using CSCT/CSDT stop criteria. Processed test data in MATLAB and performed convergence diagnostics and visualization in Python.

Wollo University

Wollo University

Desē, Ethiopia

January 2023

-

Present

Research Assistant

Civil Engineering Department

Conduct material testing of concrete, steel, and aggregates, and perform structural load tests to assess material properties and structural response. Support research on infrastructure performance under Ethiopian environmental and loading conditions, connecting experimental data with analytical models. Maintain research involvement at Wollo University alongside international studies.

Education

Polytechnic University of Catalonia

Polytechnic University of Catalonia

Barcelona, Spain

March 2026

-

September 2026

Master's Degree in Civil Engineering

Civil Engineering

University of Padua

University of Padua

Padova, Italy

October 2025

-

March 2026

Laurea Magistrale in Ingegneria Civile

Civil Engineering

Wollo University

Wollo University

Desē, Ethiopia

October 2018

-

September 2023

Bachelor of Science Degree in Civil Engineering

Civil Engineering

Skills

Skills and expertise

Structural Reliability

infrastructure assessment

Structural Analysis

Construction Technology

Software Development

Application Development

Grants and awards

Grant

EMJM Consortium Scholarship

EUR 10000

October 2025

-

September 2026

Funding agency

EMJM Consortium

Research institution

University of Padua

Secondary institutions

Polytechnic University of Catalonia, University of Minho, University of La Rochelle

Additional information

Scientific societies

American Society of Civil Engineers, Ethiopian Association of Civil Engineers

ORCID iD

0009-0009-8145-4007

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Deep-Learning Surrogate Digital Twins for Predictive Capacity Auditing and Maintenance Optimization

November 2026

DOI: 10.13140/RG.2.2.25091.26407

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Description and figures

Culverts in fast-growing cities lose capacity to siltation, yet capacity audits give a single safety factor computed from uncertain inputs. This study builds a probabilistic chain for four box culverts on the Jemo–Lancha corridor in Nifas Silk-Lafto, Addis Ababa. Rainfall (three IDF sources; 10-, 25- and 50-year storms), time of concentration, Manning roughness, runoff coefficient and a 0 to 0.5 siltation envelope were sampled with Latin Hypercube sampling and run through a dynamic-wave EPA SWMM model, giving 72,000 node results. A five-network neural ensemble was trained on these results and compared with a Gaussian process and a polynomial baseline. SWMM never flooded below a capacity factor of 1.1 and flooded in about three-quarters of the cases between 1.25 and 1.5, so the algebraic check CF > 1 is conservative. The ensemble reproduced flood volume with $R^2 = 0.998$ on held-out samples (0.984 in the extreme tail), matched SWMM failure probabilities within sampling error in 35 of 36 cases and was about 8,000 times faster. German Square was the weakest culvert and the Lancha outlet the strongest under every rainfall source. Siltation had the largest Sobol total-effect index (0.84 to 0.89), followed by time of concentration. The rainfall source moved the mean 10-year failure probability from 0.29 to 0.90 and the number of culverts needing a capacity upgrade from none to four. A maintenance matrix with a 5 % risk limit gave 10-year desilting thresholds of 6 % to 33 % blockage. Varying the assumed inlet headroom from 0 to 2 m changed failure probabilities by less than 0.01. The results are conditional on an assumed siltation range and on disagreeing rainfall sources; the code and data are open.

Figure 1. Workflow of the study, from corridor inputs through Latin Hypercube sampling, SWMM simulation, surrogate training and reliability analysis to the maintenance matrix.

Figure 2. Schematic of the Jemo-Lancha corridor from upstream (N1) to downstream (N4), with box size, reach length, slope, catchment area and runoff coefficient for each node.

Figure 7. Failure probability estimated with the surrogate (100 000 draws) against the SWMM estimate (2 000 draws, horizontal bars give the 95 % Wilson interval). Each point is one IDF source, return period and node. The dashed line is perfect agreement.

Characteristics of the four culvert nodes (registry values; Tc range is the bracket described in Section 3.3).+5

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Can natural-pozzolan geopolymer concrete beat Ethiopian PPC? A literature-calibrated assessment of carbon, cost and structural reliability

November 2026

DOI: 10.13140/RG.2.2.18380.37763

Neguse Solomon MekonnenNeguse Solomon MekonnenElsa Tareke Shibeshi

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Description and figures

Ethiopian cement is already comparatively low in carbon, so a new binder must beat Portland pozzolana cement (PPC), not ordinary Portland cement (OPC). We built a probabilistic model of embodied carbon, materials cost and column reliability for a heat-cured natural-pozzolan geopolymer concrete (GPC) against local PPC and OPC concretes. The model uses Ethiopian cement life cycle assessments, Tatek pumice chemistry, six-cube-per-age PPC C-30 strength records from an Ethiopian Roads Administration project, and dated Addis Ababa price quotes. It

puts GPC at 161 kg CO_2e/m^3 (5th to 95th percentile 135 to 201), against 303 for PPC concrete and 393 for OPC concrete. GPC costs 89% more than PPC at mid prices (67% to 115% across three PPC price cases), because the two alkali activators make up about 92% of materials cost; activator prices would have to fall to about half for cost parity. Carbon parity at column level needs only about 13 MPa mean strength, whereas cost parity cannot be reached through strength. A GPC column passes the EN 1990 target reliability index of 3.8 (β=4.8) with 4% more section area. No concrete-scale strength exists for Ethiopian natural-pozzolan GPC, and the 35 MPa reference is a paste-scale ceiling, so every GPC result is conditional on strength; the sodium silicate price unit alone can reverse the cost conclusion. Measuring concrete-scale strength and testing a locally produced pumice-derived silicate are the priorities.

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Figure 2 shows where the carbon comes from. In PPC concrete, cement and its transport account for 284 of 303 kg (94%). In the GPC the sodium silicate solution contributes 91.2 kg at production plus 10.8 kg of transport, together 102 kg or 63% of the total, NaOH 20 kg, the pozzolan 20 kg and aggregates 19 kg. The two activators together are 122 kg, 76% of the GPC footprint. Our Spearman analysis agrees: the rank correlation of sodium silicate with total GWP is 0.93, against 0.25 for the pozzolan, 0.12 for coarse aggregate, 0.10 for NaOH, 0.05 for fine aggregate and 0.00 for water (Figure 3).

Figure 3. Spearman rank correlation of each input with GPC embodied carbon.

Reported GWP reduction of geopolymer concrete against OPC concrete (compiled; contexts differ).

Strength evidence on volcanic-material binders. Rows marked † were seen as abstracts only.+5

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Thesis

Full-text available

Bayesian Updating of Bridge Structural Reliability Using Monitoring Data, Stop Criteria, and Proof Load Testing: A Case Study of the Zijlweg Viaduct

October 2026

DOI: 10.13140/RG.2.2.30377.56168

Thesis for: Master’s degree (MSc), International Master in Risk Assessment and Management of Civil Infrastructures (NORISK)Advisor: Dr. Eva Lantsoght, Dr. Rolando Chacón

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Research Interest Score

3.9

Citations

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Abstract

Aging reinforced concrete bridges across Europe face growing challenges in structural assessment, particularly where material deterioration such as alkali-silica reaction (ASR) makes analytical capacity models unreliable. Proof load testing offers a direct experimental alternative, demonstrating structural adequacy under factored loads without relying solely on analytical predictions. However, the full informational value of proof load test data remains underutilised when test results are interpreted only as binary pass or fail outcomes rather than as quantitative evidence for probabilistic reliability assessment. This dissertation applies a Bayesian reliability updating methodology to the monitoring data obtained during the full-scale proof load test of the viaduct Zijlweg, an ASR-affected reinforced concrete solid slab bridge located in Noord Brabant, the Netherlands. The viaduct was proof loaded in June 2015 at two positions, one critical for bending moment and one critical for shear, generating comprehensive sensor data including concrete strains, crack width changes, and deflections at five incremental load levels up to 1,342 kN. The Bayesian framework of De Vries et al. [1] uses these monitored values specifically concrete strain for bending and maximum weighted nominal crack width for shear to update the structural resistance distribution at each load step and calculate the annual probability of failure before, during, and after the test. In parallel, mechanics-based stop criteria derived from the Critical Shear Crack Theory (CSCT) and the Critical Shear Displacement Theory (CSDT), organised into a traffic light system, are evaluated against the same monitoring data and compared with the Bayesian reliability estimates at each load step. The corrected cross-section analysis, incorporating all four layers of bottom reinforcement from the structural drawings, gives a mean yield moment of 3,248.87 kNm with a coefficient of variation of 0.137. The posterior reliability indices after the complete proof load test are $\beta_{after}=5.33$ for bending and $\beta_{after}=4.89$ for shear, both substantially exceeding the annual target reliability for new structures ($\beta=4.7$). The annual target reliability for existing structures at the disapproval level ($\beta=4.0$) is achieved after the third load step for bending ($\beta=4.40$) and the fourth load step for shear ($\beta=4.41$). This corresponds to proof load factors of $f_{LM1}=0.866$ for bending and $f_{LM1}=0.849$ for shear. The CSCT and CSDT stop criteria indicators remained in the green light range throughout the shear test, qualitatively consistent with the positive β_{during} values at all load steps. Monitoring data provides a quantifiable advantage over the lower-bound approach: at the third shear load step, the monitoring data demonstrates $\beta_{after}=3.81$ while the lower bound gives only 1.96, an advantage of 1.85 reliability index units attributed to the crack width

information in the exponential resistance model. The findings demonstrate the practical feasibility of applying Bayesian reliability updating to real field proof load test data for an ASR-affected bridge, providing a more informative assessment than traditional analytical or lower-bound approaches. The study confirms that proof load testing, interpreted through the Bayesian framework, constitutes an effective Level IV assessment method for ASR-affected bridges that appear analytically insufficient under lower assessment levels. The stop criteria and Bayesian framework are found to be complementary rather than competing, providing operational safety signals during the test and quantitative reliability evidence after the test from the same monitoring measurements.

Public Full-text